

homologues have been cloned from *Bacillus subtilis*²³ and *Rhizobium meliloti*²⁴. An alignment of FtsZ sequences from the three bacterial species shows that the putative G1 motif (GGTGTGAA) is perfectly conserved²⁴. We suggest that FtsZ homologues in *B. subtilis* and *R. meliloti* may also be GTP-binding proteins. If true, this would imply a common signalling mechanism for the initiation of bacterial septation.

Genes encoding Ras-related GTP-binding proteins (BUD1/RSR1, refs 25, 26; CDC42, ref. 27) and a putative GDP-GTP exchange factor (BUD5, ref. 28) are necessary for bud-site selection and assembly in yeast. Moreover, CDC3, CDC10, CDC11 and CDC12 gene products, required for cytokinesis, constitute the 10-nm filament ring in the neck connecting the mother yeast cell to its bud²⁹. Our finding that FtsZ is a GTP-binding protein and the recent evidence of its localization at the septal site as a ring-like structure³⁰ suggest that the initiation of cell division in *E. coli* and budding yeast may have common features. It is conceivable that activation of FtsZ by the exchange of bound GDP by GTP, signals the ring formation at the septal site, similar to the polymerization of ATP-bound forms of actin³¹ and *E. coli* RecA³² or the GTP-bound form of tubulin³³ into filaments. Genetic evidence implicates *ftsZ* as the target of two division-inhibitors, DNA damage-inducible Sula, and MinCD, which prevents septation at old sites^{2,3}. These inhibitors could arrest division by antagonizing GTP-mediated FtsZ activation and/or signal transduction.

As with the signalling GTPases^{8,9}, binding and hydrolysis of GTP by FtsZ during the cell cycle could be regulated by specific protein ligands, such as a guanine nucleotide exchange factor and a GTPase activating protein. Analysis of the GTPase cycle of FtsZ may reveal shared features between the primordial division machinery in *E. coli* and the eukaryotic cell division apparatus, and may offer insights into bacterial cell division. □

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The essential bacterial cell-division protein FtsZ is a GTPase

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CYTOKINESIS defines the last stage in the division cycle, in which cell constriction leads to the formation of daughter cells. The biochemical mechanisms responsible for this process are poorly understood. In bacteria, the *ftsZ* gene product, FtsZ, is required for cell division^{1–3,17}, playing a prominent role in cytokinesis. The cellular concentration of FtsZ regulates the frequency of division⁴ and genetic studies have indicated that it is the target of several endogenous division inhibitors⁵. At the time of onset of septal invagination, the FtsZ protein is recruited from the cytoplasm to the division site, where it assembles into a ring that remains associated with the leading edge of the invaginating septum until septation is completed⁶. Here we report that FtsZ specifically binds and hydrolyses GTP. The reaction can be dissociated into a GTP-dependent activation stage that is markedly affected by the concentration of FtsZ, and a hydrolysis stage in which GTP is hydrolysed to GDP. The results indicate that GTP binding and hydrolysis are important in enabling FtsZ to support bacterial cytokinesis, either by facilitating the assembly of the FtsZ ring and/or by catalysing an essential step in the cytokinetic process itself.

The FtsZ peptides from the three bacterial species of known sequence^{7–9} share a 13 amino-acid segment that is completely conserved and that includes a 7 amino-acid stretch almost identical to the most highly conserved sequence among the α -, β - and γ -tubulins of eukaryotic cells (Fig. 1). It is proposed that this tubulin signature sequence¹⁰ is involved in the binding of GTP by tubulins^{11,12}, prompting us to investigate whether FtsZ might be a GTPase.

Evidence that FtsZ is a GTP-binding protein was obtained from experiments in which FtsZ was specifically crosslinked to [α -³²P]GTP by ultraviolet irradiation (Fig. 2). When crude *Escherichia coli* cell extracts containing elevated concentrations of endogenously generated FtsZ were subjected to the crosslinking procedure, SDS-gel electrophoresis revealed a labelled band whose position corresponded to that of FtsZ. The extent of labelling paralleled the concentration of FtsZ in the extracts (Fig. 2a, b, lanes 1–4). The ability to bind GTP was retained by the purified protein (Fig. 2a, b, lane 5) and did not require Mg²⁺ (Fig. 2c, lane 1). The nucleotide-binding reaction was specific for GTP and GDP, as shown by competition experiments with nonradioactive nucleotides (Fig. 2c, lanes 2–6) and by the inability to crosslink [α -³²P]ATP to FtsZ (data not shown).

A single amino-acid substitution within the tubulin signature sequence of FtsZ (FtsZ84; Fig. 1) leads to a failure to initiate septum formation in cells grown at elevated temperatures. As shown in Fig. 2b (lanes 6, 7), the efficiency of crosslinking of [α -³²P]GTP to the mutant FtsZ84 protein was significantly lower



FIG. 1 Tubulin signature sequence in FtsZ. The stretch of 13 amino acids that is completely conserved among FtsZ peptides is shown for the three bacterial species for which the sequence is known (*Escherichia coli*⁷, *Bacillus subtilis*⁸ and *Rhizobium meliloti*⁹). The substitution of serine for glycine in the mutant FtsZ84 protein of *E. coli*¹⁴ leads to a failure in initiating cytokinesis when mutant cells are grown at elevated temperatures¹⁵. The lower portion of the figure shows the 7 amino-acid tubulin signature sequence¹⁰ that is conserved among tubulins of higher organisms.

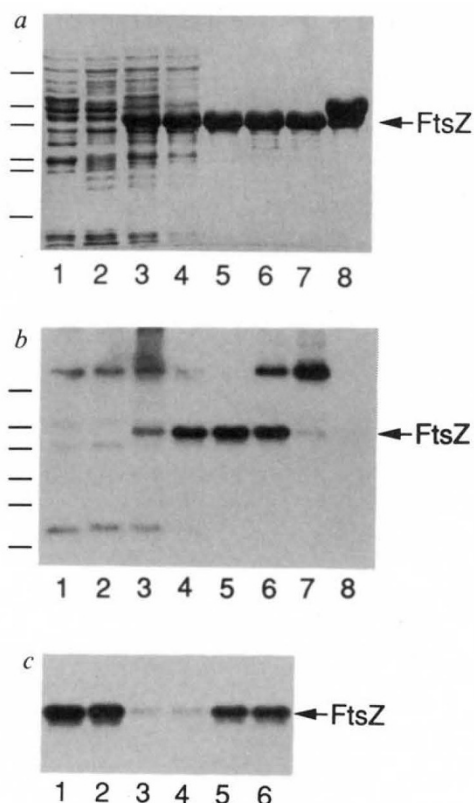


FIG. 2 Covalent crosslinking of $[\alpha\text{-}^{32}\text{P}]\text{GTP}$ to FtsZ. **a**, Coomassie blue-stained SDS gels showing the protein profiles of the samples used in this study. Lanes 1–4, S200 extracts of strains with different levels of *ftsZ* expression: lane 1, PB115/pBSKS[pBluescriptKS vector]; lanes 2–4, PB115/pDB189[P_{lac}::ftsZ]. Cells were grown in LB medium containing 0.2% glucose to repress baseline *ftsZ* expression (lane 2), in LB medium (lane 3) or in LB medium containing 0.84 mM isopropyl- β -D-thiogalactoside (IPTG) (lanes 1, 4). Lane 5, purified FtsZ protein. Lane 6, partially purified FtsZ protein. Lane 7, partially purified FtsZ84 protein. 50 μg protein was applied to lanes 1–4 and 30 μg to lanes 5–7. Lane 8, 60 μg chicken ovalbumin. The positions of protein molecular mass standards (66, 45, 36, 29, 24 and 20K, from top to bottom) are indicated to the left of the gel. **b**, The samples shown in the corresponding lanes of **a** were added to 0.5 μM $[\alpha\text{-}^{32}\text{P}]\text{GTP}$ (400 Ci mmol⁻¹), 20 mM Tris-acetate (pH 7), 100 mM potassium acetate, 1 mM EDTA, 0.5 mM dithiothreitol (DTT) and 10 mM magnesium acetate, then immediately irradiated with ultraviolet light for 30 min at 4 °C as described¹⁶ before analysis by SDS-gel electrophoresis and autoradiography. The gel lanes contained 7.1 μg (lanes 1–4), 5.7 μg (lane 5), 2.9 μg (lanes 6, 7) and 10 μg (lane 8) of protein, respectively. Lanes 6 and 7 show that a minor 75K contaminant peptide, which very efficiently forms photoadducts with $[\alpha\text{-}^{32}\text{P}]\text{GTP}$, is present in the partially purified preparations of both FtsZ (lane 6) and FtsZ84 (lane 7). This protein is not present in the more purified preparation of FtsZ (lane 5). **c**, Purified FtsZ (lane 5 of panel **a**) was irradiated with ultraviolet light in the presence of $[\alpha\text{-}^{32}\text{P}]\text{GTP}$ as described in **b**. Lane 1, Mg^{2+} was omitted. Lane 2, complete reaction mixture. Lanes 3–6, the following unlabelled nucleotides (50 μM) were added to the complete reaction before crosslinking: lane 3, GTP; lane 4, GDP; lane 5, GMP; lane 6, ATP. Addition of non-radioactive dGTP, CTP or UTP gave results identical to those shown in lanes 5 and 6.

METHODS. Purified FtsZ was prepared from cells of an overproducing *E. coli* strain (PB115/pDB189) grown at 37 °C in LB medium containing 0.84 mM IPTG. Cells were broken in a French pressure cell, centrifuged for 180 min at 200,000g, the supernatant (S200) was fractionated by precipitation with ammonium sulphate, followed by column chromatography on Q-Sepharose, HA-Ultragel and Superose-12. Partially purified FtsZ and FtsZ84 proteins were prepared from the S200 fraction of PB115/pDB189[P_{lac}::ftsZ] and PB115/pDB254[P_{lac}::ftsZ84] cells grown at 30 °C by precipitation with ammonium sulphate, followed by column chromatography on Superose-12. All parameters during the preparation of partially purified FtsZ and FtsZ84 were identical. The *ftsZ84* allele in pDB254 was obtained by homologous recombination between pCX41 (*ftsZ*⁺) and the chromosome of strain AW40, *ftsZ84*, (ref. 3).

than that of the wild-type protein. Thus a single mutation affected both the biological function and the GTP-binding ability of the protein.

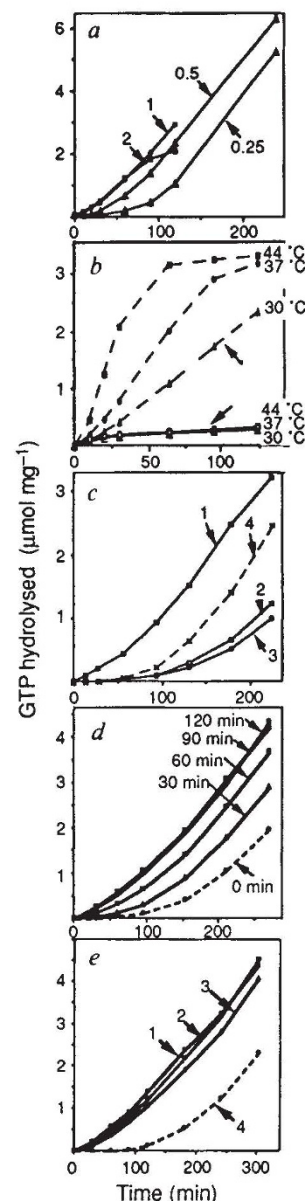
Evidence that FtsZ is a GTPase was obtained by measuring the ability of the purified protein to convert $[\alpha\text{-}^{32}\text{P}]\text{GTP}$ to $[\alpha\text{-}^{32}\text{P}]\text{GDP}$ (Fig. 3a). In contrast to the binding of GTP to FtsZ, already described, FtsZ-mediated hydrolysis of GTP required the presence of Mg^{2+} .

A comparison of mutant and wild-type proteins revealed that the GTPase activity of the mutant FtsZ84 protein was significantly lower than that of wild-type FtsZ (Fig. 3b). The mutant protein also failed to show the normal temperature-dependent increase in activity that is seen with the wild-type enzyme.

The GTPase activity of FtsZ showed an unusual dependence on protein concentration. As shown in Fig. 3a, a lag period occurred before the onset of significant GTP hydrolysis, the duration of which was inversely related to FtsZ concentration.

FIG. 3 GTPase activity of FtsZ. **a**, GTPase activity at different concentrations of FtsZ. Purified FtsZ (see Fig. 2a, lane 5) was assayed at 37 °C for GTPase activity at 0.25, 0.5, 1 and 2 mg ml⁻¹ protein. The reaction mixtures were equilibrated at 37 °C for 5 min before starting the reaction by the addition of Mg^{2+} . **b**, Mutant compared with wild-type FtsZ. Assays were done on partially purified FtsZ and FtsZ84 proteins (described in Fig. 2a, lanes 6 and 7) at 44 °C, 37 °C and 30 °C. The reactions were started by addition of protein (1 mg ml⁻¹) to the other components of the reaction mixture that had been preincubated for 10 min at the indicated temperatures. Dashed lines, FtsZ; solid lines, FtsZ84. **c**, Effect of preincubation. Purified FtsZ (0.3 mg ml⁻¹) was preincubated at 37 °C for 80 min in reaction mixtures from which Mg^{2+} (reaction 1), GTP (reaction 2) or both Mg^{2+} and GTP (reaction 3) were omitted. The reactions were then started (at time 0) by addition of the missing components. There was no detectable hydrolysis of GTP during the preincubation with $[\alpha\text{-}^{32}\text{P}]\text{GTP}$ in the absence of Mg^{2+} (reaction 1). Reaction 4 was started by addition of FtsZ to the other components of the reaction mixture that had been preincubated for 80 min. **d**, Time course of activation. Purified FtsZ was incubated with GTP before addition of Mg^{2+} as described for reaction 1 in **c**. The time of preincubation varied from 0 to 120 min as indicated in the figure. **e**, Effect of GTP concentration during preincubation. Purified FtsZ was incubated with GTP for 90 min, as described for reaction 1 in **c**. The concentrations of GTP present during the preincubation period were 5 mM (reaction 1), 1 mM (reaction 2), 0.5 mM (reaction 3), or 0 mM (reaction 4). After the preincubation period the reactions were started (time 0) by addition of GTP to 5 mM, and magnesium acetate to 15 mM.

METHODS. For GTPase assays the complete reaction mixture contained 5 mM $[\alpha\text{-}^{32}\text{P}]\text{GTP}$ (40 mCi mmol⁻¹), 15 mM magnesium acetate, reaction buffer (40 mM Tris-acetate (pH 7), 200 mM potassium acetate, 2 mM EDTA, 1 mM DTT and 0.5% Triton X-100) and protein as indicated. At intervals, aliquots from the reaction mixtures were subjected to thin layer chromatography and the amount of GTP converted to GDP was determined¹⁶.



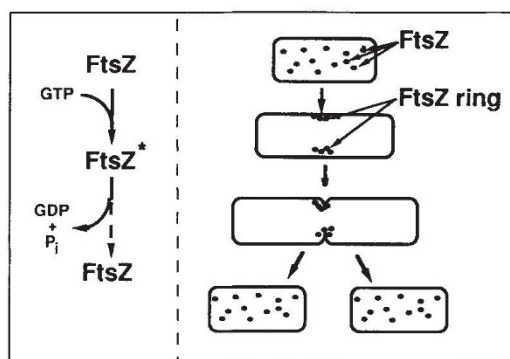


FIG. 4 *In vitro* properties and *in vivo* behaviour of FtsZ. The left panel indicates the two stages in the *in vitro* FtsZ GTPase reaction as described in the text. The right panel indicates the *in vivo* behaviour of FtsZ during the cell cycle⁶. FtsZ* indicates the activated form of FtsZ.

With longer incubation period, the specific activity of the FtsZ GTPase reached the same level at all protein concentrations as shown by the parallel slopes of the activity curves in the time-course plots. We conclude that during the lag period an activation step occurred, leading to formation of a fully functional enzyme, and that the duration of the activation step was significantly affected by protein concentration.

To determine which component of the reaction mixture was required for the activation step, purified FtsZ was preincubated with the different individual components of the mixture before the reaction was started by addition of the missing ingredients. As shown in Fig. 3c, preincubation with GTP resulted in virtual loss of the lag period whereas preincubation with the other components of the reaction mixture failed to do so. Preincubation for 90 min was required for maximal activation with 5 mM GTP (Fig. 3d). There was no measurable hydrolysis of GTP during the preincubation. These results define two stages in the GTPase reaction: an activation stage that requires binding of the nucleotide and whose duration is strongly dependent on protein concentration, and a Mg^{2+} -dependent catalytic stage in which GTP is hydrolysed to GDP.

A 10-fold decrease in GTP concentration during the preincubation, which should greatly decrease the rate of interaction of GTP with FtsZ, did not significantly affect the extent of activation, as measured by suppression of the lag period (Fig. 3e). This suggests that the binding of FtsZ to GTP is not rate-limiting in formation of the active enzyme. In contrast, as shown in Fig. 3a, small changes in protein concentration had a marked effect on the time needed to activate the enzyme, as indicated by the length of the lag.

This behaviour resembles that of self-associating systems such as the nucleotide-dependent assembly of microtubules and actin filaments¹³, and indicates that formation of the active FtsZ GTPase is dependent on interactions between FtsZ subunits, perhaps involving the assembly of FtsZ into a multimeric complex. In gel filtration studies, purified FtsZ behaved as a series of oligomers ranging in size from 6-mers to 25-mers, with a mean M_r of 400 K (10-mer) (data not shown). Whether binding of GTP alters the oligomeric state of FtsZ remains to be tested directly.

Recent immunolocalization studies⁶ defined several stages in the behaviour of FtsZ during the division cycle (Fig. 4). In the first stage, FtsZ moves from its normal cytoplasmic location to assemble at the inner surface of the cytoplasmic membrane into a ring that extends circumferentially around the cell at the site of division. This is followed by the cytokinesis stage, during

which FtsZ remains at the leading edge of the invaginating septum. When septation is completed the protein disappears from this site and is once again found in the cytoplasm.

It is likely that the GTPase activity of FtsZ plays a role in one or more of these stages. It is tempting to speculate that the concentration-dependent activation step, induced by GTP binding in the *in vitro* system, may be related to a multimerization step that leads to production of the FtsZ ring *in vivo*. The subsequent hydrolysis of GTP may play a role in the cytokinetic event itself and/or in the release of FtsZ that occurs after septation is completed. The assignment of a biochemical activity to FtsZ should permit an *in vitro* approach to studying aspects of the division process which have previously been inaccessible. □

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